

Interferon α -2b gains high sustained response therapy for advanced essential thrombocythemia and polycythemia vera with JAK2V617F positive mutation



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ARTICLE INFO

Article history:

Received 8 May 2014

Received in revised form 20 June 2014

Accepted 28 June 2014

Available online 15 July 2014

Keywords:

Interferon

Essential thrombocythemia

Polycythemia vera

JAK2V617F mutation

ABSTRACT

Objective: This open-label, prospective, observational study aimed to evaluate treatment response, efficacy therapy and safety to IFN α -2b for the essential thrombocythemia (ET) and polycythemia vera (PV) with JAK2V617F positive mutation.

Method: A total of 123 ET patients received IFN α -2b therapy with JAK2V617F positive or negative mutation; and 136 PV patients with JAK2V617F⁺ received IFN α -2b or hydroxyurea (HU) therapy according to random number assignment (ages 18–65 years old).

Result: ET patients receiving IFN α -2b with JAK2V617F⁺ had a greater advantage in overall hematologic response (OHR) than JAK2V617F⁻ (83.3% versus 61.4%, $P < 0.01$). For PV patients with JAK2V617F⁺, IFN had no OHR superiority to HU (70.3% versus 70.8%, $P > 0.05$), but which gained a greater satisfactory molecular response than HU (54.7% versus 19.4%, $P < 0.01$). IFN significantly decreased the phlebotomy rate, which was better than HU for MPDs patients with OHR than HU (3.6% versus 65.7%, $P < 0.01$). Furthermore, ET patients with JAK2V617F⁺ demonstrated a definite advantage over JAK2V617F⁻ in five-year PFS (75.9% versus 47.6%, $P < 0.05$). For PV patients with JAK2V617F⁺, IFN α -2b was superior to HU in five-year PFS (66.3% versus 46.7%, $P < 0.01$). Moreover, IFN α -2b also contributed to improved vasomotor symptoms in MPDs, and especially significantly decreased the incidence of distal paresthesias (14.1% versus 37.5%) and erythromelalgia (9.4% versus 29.2%) better than HU ($P < 0.01$). Meanwhile, IFN did not observe the severe hematological adverse events in patients with PV or ET.

Conclusion: The data confirmed that IFN α -2b benefited the patients with ET or PV, particularly for JAK2V617F⁺ mutation.

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1. Introduction

Chronic myeloproliferative disorders (MPDs) constitute one of five categories of myeloid malignancies, according to the World Health Organization (WHO) classification system for hematopoietic tumors [1]. Essential thrombocythemia (ET) and polycythemia vera (PV) are sub-categories of MPDs [2]. For a long time, the absence of specific genetic or molecular markers caused problems in the differential diagnosis in MPDs. Recently, the acquired gain-of-function

V617F mutation in the Janus Kinase 2 (JAK2V617F) has been found to be the major mutation involved in BCR/ABL negative classical MPDs. It is present in the mutational frequency of 96% in PV, 55% of ET patients [3,4].

The risk of thrombosis, in both ET and PV, exceeds 20% and a substantial proportion of patients experience vasomotor disturbances [5]. Risk factors for shortened survival in both ET and PV include history of thrombosis, leukocytosis, advanced age and anemia. Accordingly, the goal of current therapy in PV and ET is primarily to prevent thrombohemorrhagic complications.

The related treatment for patients with ET and PV should aim at avoiding thromboembolic complications by bone marrow-suppressive therapy. Such treatments ideally should not increase the risk of acute leukemia [6]. Hydroxyurea (HU) generally is

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Table 1
Characteristics of enrolled patients with MPDs.

ET arm with IFN α -2b	JAK2V617F ⁺ n = 66		JAK2V617F ⁻ n = 57		P value
	Median	Range	Median	Range	
Age	47	26–65	50	26–64	>0.05
WBC $\times 10^9$ /L	26	4.7–47.6	22	3.0–48.6	>0.05
Hemoglobin, g/L	161	115–184	166	119–189	>0.05
Platelets	1125	960–1400	1127	998–1474	>0.05
PV arm with JAK2V617F ⁺	IFN α -2b n = 64		Hydroxyurea n = 72		P value
	Median	Range	Median	Range	
Age	47	26–64	47	29–64	>0.05
WBC $\times 10^9$ /L	26.1	6.6–50.1	26	7.2–57.9	>0.05
Hemoglobin, g/L	197	185–213	196	188–219	>0.05
Platelets	600	423–900	623	426–936	>0.05

considered the treatment of choice for MPDs-associated thrombocythemia [7] and recently showed superior activity over anagrelide. On the other hand, the clinical effectiveness of interferon α -2b (IFN α -2b) had been demonstrated nearly a quarter of a century ago in PV in the US, and almost simultaneously in ET patients by French and Austrian groups [8].

At present, it is still not determined whether IFN α -2b gains the therapy superiority for the PV and ET with or without positive JAK2V617F mutation. The study represents the multicenter study reviews to evaluate efficacy and adverse events of IFN α -2b for advanced ET or PV adult patients (hematocrit >45% or extreme thrombocytosis $>1000 \times 10^9$ /L) or thrombosis history related to JAK2V617F mutation.

2. Methods

2.1. Study population

This open-label, observational, multicenter clinical assessment enrolled participants from February 2009 to March 2014. All adult patients (ages 18–65 years old) were diagnosed according to the WHO 2008 classification and fit a composite assessment of clinical and laboratory features [9]. Enrolled advanced patients with extreme thrombocytosis risks or thrombosis history gave written informed consent before entering the study. This study was approved by the institution review board at each of the participating centers [10]. Eligible patients received treatment schedules in different sites based on a random number assigned.

ET patients with JAK2V617F positive or negative mutation only received interferon α -2b regimen. Because patients with PV had high JAK2V617F positive rates (over 90%), only patients with JAK2V617F⁺ mutations were included in the study; these patients respectively received IFN α -2b or HU regimen (Fig. 1).

In the PV arm, 41/66 (62.1%) patients with JAK2V617F⁺ accomplished full course IFN treatment, and 35/57 (61.4%) patients with JAK2V617F⁻ finished IFN treatment until the termination of the study. In the ET group, two patients were not tolerant of IFN in the first week and joined the HU treatment arm. 49/64 patients received the IFN-full course treatment (Fig. 1). A total of 50/123 patients with ET and 17/136 patients with PV did not persist on a full-course treatment (Fig. 1).

Bone marrow (BM) was aspirated for morphological, biopsy and molecular analysis. JAK2V617F was detected in DNA samples extracted from BM specimens by polymerase chain reaction and quantitated by a pyrosequencing assay.

2.2. Selection criteria

(1) Patients with a diagnosis of advanced ET or PV (hematocrit >45% or extreme thrombocytosis $>1000 \times 10^9$ /L) either newly diagnosed or previously treated; (2) Patients with extreme thrombocytosis risks or thrombosis history; (3) The Karnofsky performance status (KPS) scores were at least 60; (4) A serum total bilirubin concentration was found to be no higher than twice the upper limit of the normal range; (5) A serum concentration of alanine aminotransferase was no higher than 2.5 times the upper limit of the normal range; (6) Patients with symptoms of congestive heart failure were excluded.

2.3. Treatment schedule

IFN α -2b therapy regimen: original dose was three million units SC three times a week for two years, and maintenance dose was five million units SC two times a

week until the termination of the study. HU therapy regimen: the starting dose of HU is 15–20 mg/kg/day until response is obtained, the maintenance dose is titrated to keep platelet count in the normal range and leukocyte count $>2.5 \times 10^9$ /L [11] until the termination of the study.

Low-dose aspirin (100 mg daily) treatment was received by enrolled ET and PV patients had extreme thrombocytosis risks or thrombosis history for supportive therapy. Patients with PV (hematocrit >50%) received phlebotomy treatment when they received IFN or HU in order to alleviate vasomotor symptoms.

2.4. Study endpoints

Complete hematologic response (CHR) in patients with ET was defined as normalization of platelet counts ($\leq 400 \times 10^9$ /L) without thromboembolic events. Partial hematologic response (PHR) required at least a 50% reduction in platelet counts (but still $>400 \times 10^9$ /L). CHR in patients with PV was defined as normalization of hematocrit (<45% in males and <42% in females), WBC and platelet counts, and spleen size, without the absence of thromboembolic events. PHR required the documentation of at least a 50% reduction in phlebotomy requirements or spleen size [12].

Complete molecular response (CMR) required undetectable levels of JAK2V617F mutation; partial molecular response (PMR) required $\geq 50\%$ reduction of baseline JAK2V617F mutation level; and minor molecular response (MMR) required 20% to 49% reduction of baseline JAK2V617F mutation level [11,13].

Progression-free survival (PFS) was defined as the time from the diagnosis to progression, including thrombosis, bleeding, spleen enlargement, severe myelofibrosis or death from any cause in the treatment. Overall response (OS) was measured from diagnosis until death from any cause. Patients who were still alive at the date of last contact were then censored. Adverse drug events were monitored and recorded according to the Common Terminology Criteria for Adverse Events (CTCAE) v3.0 (National Cancer Institute, 2003) at each visit after the end of the consolidation treatment.

2.5. Statistical analysis

Statistical analyses were performed with SPSS 13.0 statistical software. The features of patients (stratifications of WBC, hemoglobin, platelets and age distribution) were evaluated by *t*-test; complete response, phlebotomy rate, vasomotor symptoms and adverse-event response were evaluated using a Chi-square test; PFS and OS were estimated using the Kaplan–Meier method. A *P* value of <0.05 was considered statistically significant.

3. Results

3.1. Enrolled patients with clinical characteristics

The median age of 123 patients with ET was 49 years old. Two treatment groups JAK2V617F⁺ versus JAK2V617F⁻ were well balanced in age (median: 47 versus 50 years old), WBC (median: 26 versus 22×10^9 /L), hemoglobin (median: 161 versus 166 g/L), and platelet (median: 1125 versus 1127×10^9 /L) counts (*P* > 0.05) at the time of random assignment (Table 1).

The median age of 136 PV patients with JAK2V617F⁺ was 47 years old. Two treatment groups (IFN α -2b versus HU) were well balanced in age (median: 47 versus 47 years old), WBC (median: 26.1 versus 26×10^9 /L), hemoglobin (median: 197 versus 196 g/L),

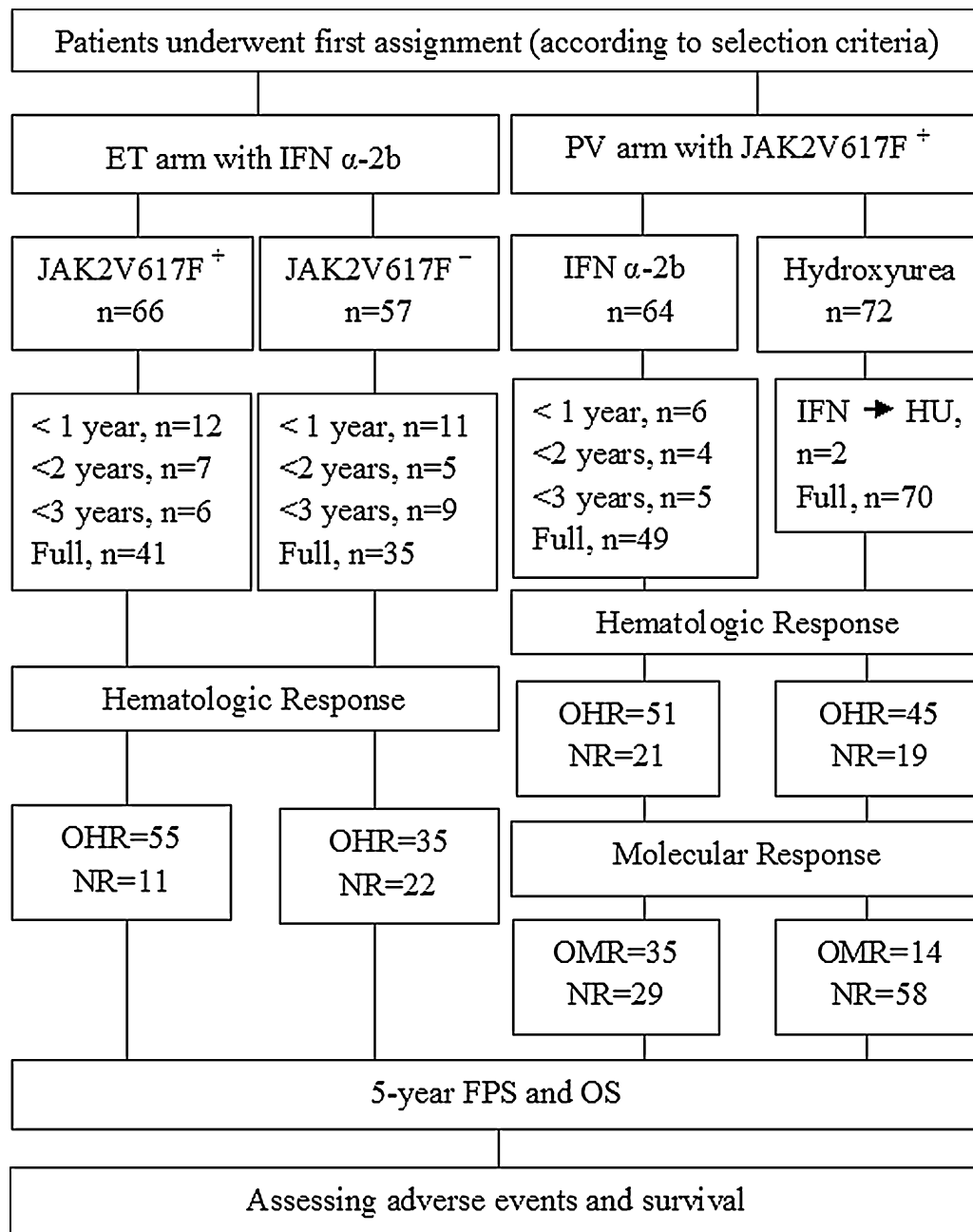


Fig. 1. Study design and patients disposition. OHR, overall hematologic response; OMR, overall molecular response.

and platelet (median: 600 versus $623 \times 10^9/L$) counts risk group ($P > 0.05$) at the time of the assignment (Table 1).

3.2. Hematologic and molecular response to IFN α -2b

JAK2V617F was detected in 66 of the 123 ET patients treated with IFN α -2b. The overall hematologic response (OHR) rate was 83.3% for ET with JAK2V617F⁺ and 61.4% for ET with JAK2V617F⁻ ($P < 0.01$). Accordingly, the OHR rate was 70.3% for receiving IFN α -2b and 70.8% for receiving HU in PV patients with JAK2V617F⁺ ($P > 0.05$, Table 2). Molecular monitor was achieved in PV patients with JAK2V617F-positive in the termination of IFN. Unlike a hematologic response, a molecular response was more highly achieved in PV patients receiving IFN α -2b than HU. The overall molecular response (OMR) rate was 54.7% for ET and 19.4% for PV ($P < 0.01$, Table 3).

3.3. Phlebotomy rate to PV after treatment

For patients with OHR, only 3.6% (2/55) of patients receiving IFN needed to receive phlebotomy, but 65.7% (23/35) of patients receiving HU depended on phlebotomy treatment ($P < 0.01$). Certainly for patients without OHR, most patients (87.9%) still had to receive phlebotomy with either IFN or HU ($P > 0.05$, Table 4).

3.4. Progression-free and overall survival

We analyzed five-year PFS and OS in ET and PV arm. Significant differences were found in five-year PFS between JAK2V617F⁺ and JAK2V617F⁻ group for ET patients, 75.9% versus 47.6% ($P < 0.05$, Fig. 2A); but with no differences in five-year OS between them, 94.6% versus 93.2% ($P > 0.05$, Fig. 2B). Similarly, patients reaching

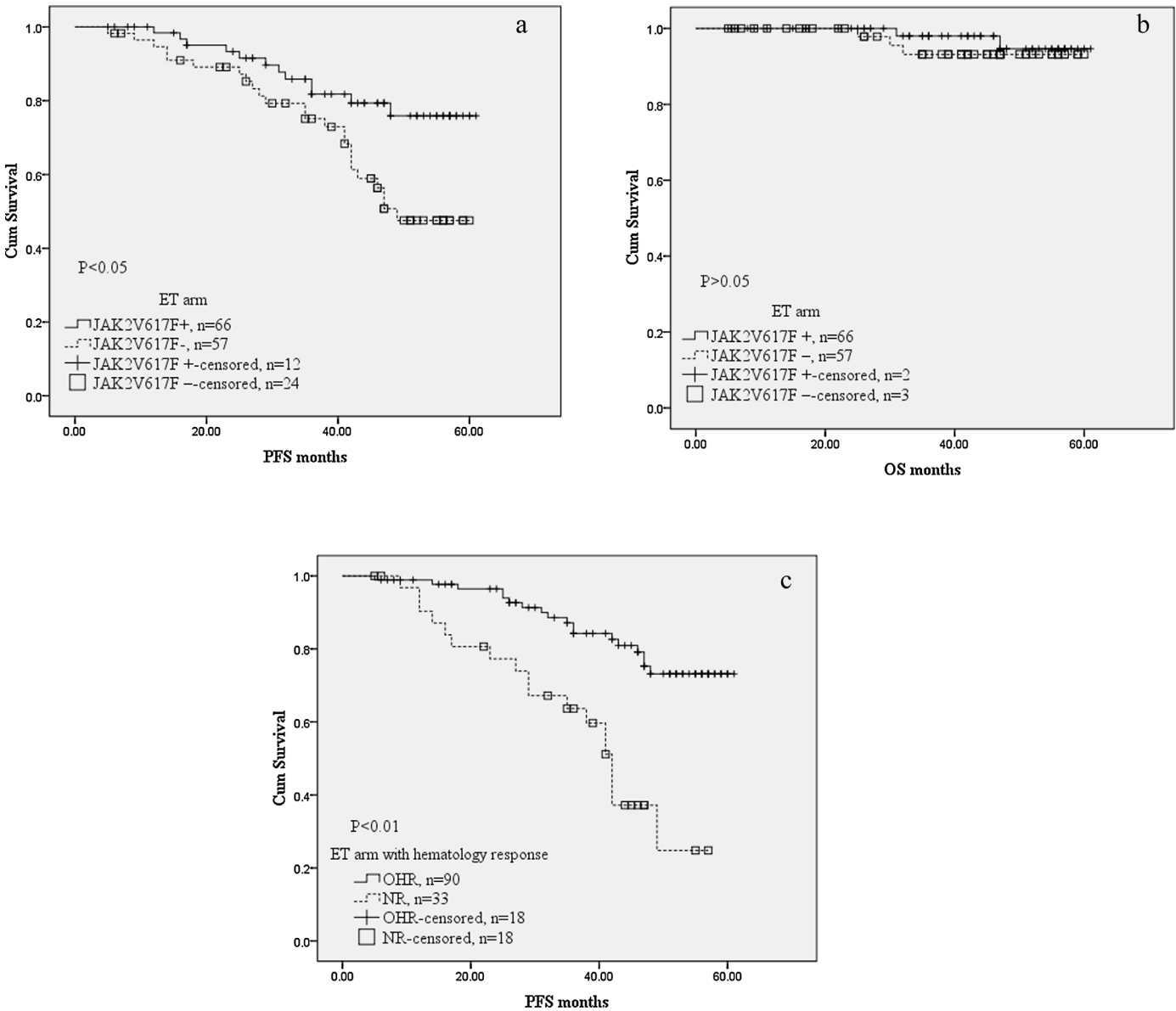


Fig. 2. (A) PFS by ET arm. (B) OS by ET arm. (C) PFS by ET arm with hematology response. OHR, overall hematologic response; NR, no OHR.

Table 2
Hematologic Response to ET and PV.

ET arm with IFN α-2b	JAK2V617F ⁺ n = 66		JAK2V617F ⁻ n = 57		P value
	n	(%)	n	(%)	
CHR	23		12		<0.01
PHR	32		23		
OHR	55	83.3	35	61.4	
PV arm with JAK2V617F ⁺	IFN α-2b n = 64		Hydroxyurea n = 72		P value
	n	(%)	n	(%)	
CHR	19		22		>0.05
PHR	26		29		
OHR	45	70.3	51	70.8	

Abbreviations: CHR, complete hematologic response; PHR, partial hematologic response; MHR, minor hematologic response; OHR, overall hematologic response; ET, essential thrombocythemia.

OHR had more PFS advantages than patients who had not gained OHR (73.2% versus 24.8%, $P < 0.01$, Fig. 2C).

For PV patients, significant differences were found in five-year PFS between IFN α-2b and HU, 66.3% versus 46.7% ($P < 0.01$, Fig. 3A); however, with no differences in 5-year OS between IFN α-2b and HU for PV group, 94.3% versus 94.1% ($P > 0.05$, Fig. 3B). Further PFS analysis indicated no differences between patients who had or had

Table 3
Molecular Response to PV arm with JAK2V617F⁺.

PV arm with JAK2V617F ⁺	Interferon α-2b n = 64		Hydroxyurea n = 72		P value
	n	(%)	n	(%)	
CMR	5		0		<0.01
PMR	21		5		
MMR	9		9		
OMR	35	54.7	14	19.4	

Abbreviations: CMR, complete molecular response; PMR, partial molecular response; MMR, minor molecular response; OMR, overall molecular response; PV, polycythemia vera.

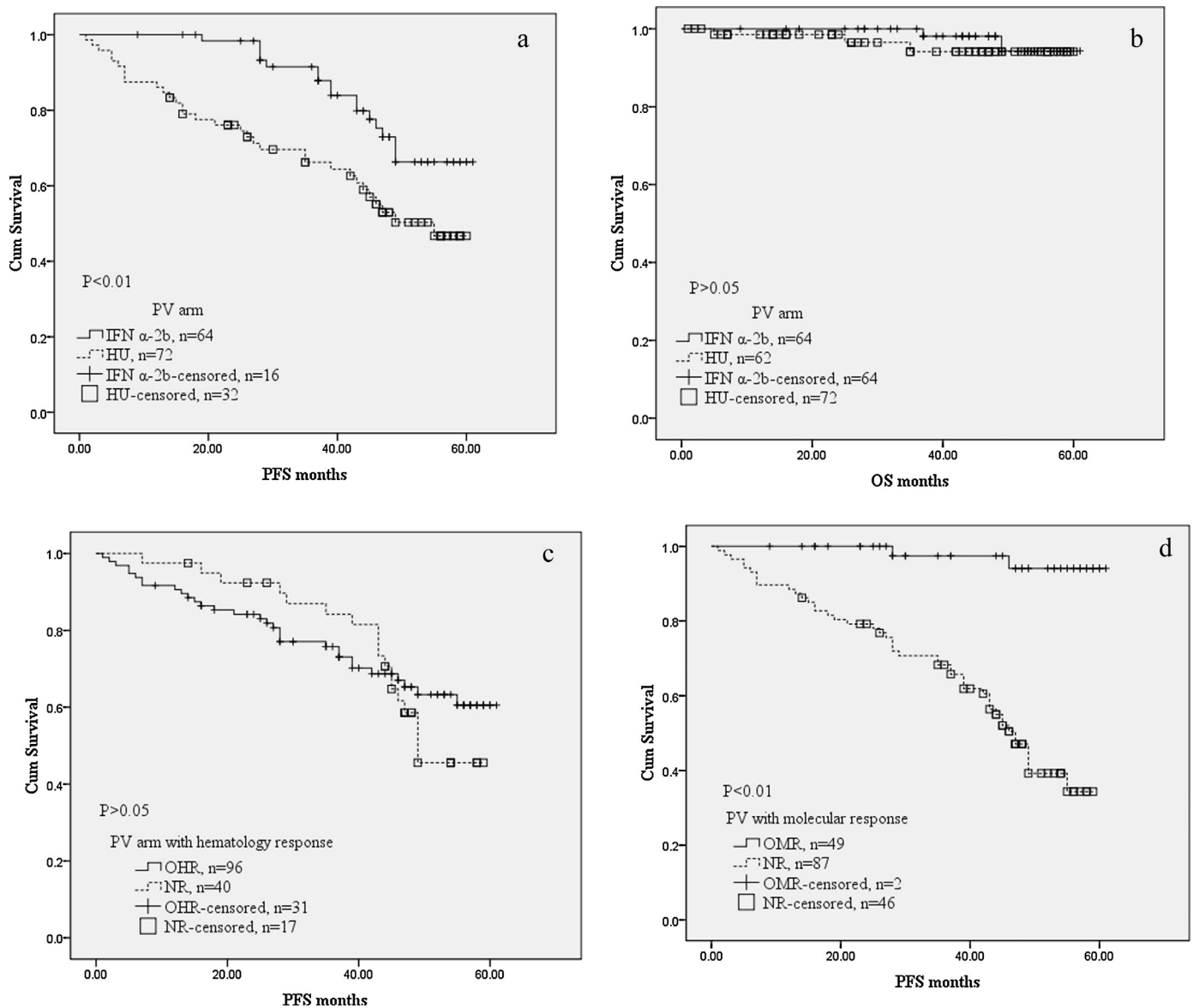


Fig. 3. (A) PFS by PV arm. (B) OS by PV arm. (C) PFS by PV arm with hematology response. OHR, overall hematologic response; NR, no OHR. (D) PFS by PV arm with molecular response. OMR, overall molecular response; NR, no OMR.

not gained OHR (60.5% versus 45.6%, $P > 0.05$, Fig. 3C). However, patients reaching OMR had more PFS advantages than patients with no OMR (94.1% versus 34.3%, $P < 0.01$, Fig. 3D).

3.5. Vasomotor symptom improvements to ET and PV after treatment

Patients with MPDs receiving either IFN α -2b or HU had obvious improvements of vasomotor symptoms. Only 0.39% of patients had atypical chest, 0.77% patients had pain, 3.1% patients had light

headedness, and 4.2% patients had visual disturbances symptoms after IFN or HU treatment (Table 5). Moreover, IFN ET patients were more likely to see improvement in headaches, distal paresthesias and erythromelalgia than PV patients ($P < 0.01$). Meanwhile, IFN also had significant advantages in improving distal paresthesias (14.1% versus 37.5%) and erythromelalgia (9.4% versus 29.2%) than HU in the PV group ($P < 0.01$, Table 5).

3.6. Adverse events

The total deaths of 5/123 (4.1%) patients with ET and 5/136 (3.7%) patients with PV were recorded during follow up. Among them, seven patients received IFN and another three patients received HU treatment. 66/123 of ET patients and 78/136 of PV patients developed some toxicity, but this was only generally Grade one or two. The main mild adverse event was fever ($>38^{\circ}\text{C}$), which was tolerated by most patients (Table 6).

For ET patients receiving IFN α -2b therapy, low rates of Grade three or four toxicities, including infection (0.8%), allergic reaction (2.4%), nausea (1.6%), dermatology/skin (0.8%), general cardiac

Table 4
Phlebotomy rate to PV after treatment.

Phlebotomy rate	IFN α -2b N = 66		Hydroxyurea N = 57		P value
	n	(%)	n	(%)	
OHR	2/55	3.6	23/35	65.7	<0.01
NR	9/11	81.1	20/22	90.9	

Abbreviations: OHR, overall hematologic response; NR, non-OHR.

Table 5

Vasomotor symptoms improvements to ET and PV after treatment.

	ET with IFN α -2b		PV with IFN α -2b		PV with HU		P value
	n	(%)	n	(%)	n	(%)	
Total	123		64		72		
Headaches	9	7.3	17	26.6	19	26.4	<0.01
Visual disturbances	6	4.9	2	3.1	3	4.2	>0.05
Light headedness	2	1.6	3	4.7	3	4.2	>0.05
Pain	1	0.8	0	0	1	1.4	>0.05
Distal paresthesias	8	6.5	9	14.1	27	37.5	<0.01
Erythromelalgia	2	1.6	6	9.4	21	29.2	<0.01
Atypical chest	1	0.8	0	0	0	0	>0.05

Table 6

Main adverse events for ET and PV arm (grade 3 or 4).

Grade 3 or 4	ET arm		PV arm		HU		P value
	n	(%)	n	(%)	n	(%)	
Total patients	123		64		72		
Leukocytes	0	0	0	0	5	6.9	<0.05
Platelets	0	0	0	0	1	1.4	>0.05
Infection	1	0.8	0	0	1	1.4	>0.05
Allergic reaction	3	2.4	1	1.6	0	0	>0.05
Nausea	2	1.6	1	1.6	7	9.7	<0.05
Dermatology/skin	1	0.8	2	3.1	1	1.4	>0.05
Cardiac general	2	1.6	1	1.6	0	0	>0.05
Liver dysfunction	2	1.6	2	3.1	4	5.6	>0.05
Renal failure	0	0	0	0	0	0	>0.05

(1.6%) and elevated liver function tests (1.6%) were found. No renal failure was found in ET patients.

Furthermore, for PV patients with JAK2V617F⁺ mutation, no neutropenia, infection, thrombocytopenia or renal failure were observed in the IFN α -2b therapy group. Significant differences in neutropenia and nausea Grade three-four between IFN α -2b and HU therapy ($P < 0.05$) were found. Other AEs Grade three-four were observed in them, including liver dysfunction 3.1% versus 5.6% ($P > 0.05$); dermatology/skin 3.1% versus 1.4% ($P > 0.05$). Only one patient receiving IFN α -2b had an allergic reaction (Grade three or four, 1.6%); no allergic reaction in patients receiving HU therapy ($P > 0.05$) was observed.

4. Discussion

PV and ET are chronic MPDs characterized by clonal expansion of an abnormal hematopoietic stem/progenitor cell. Patients with MPDs have a clinical course characterized by a thrombophilic state, which results in arterial and venous thromboses that greatly impacts morbidity and mortality. For ET and PV patients with extreme thrombocytosis risks or thrombosis history, the agent of choice to manage hyperviscosity is HU or IFN. HU provides effective control of the hematocrit in approximately 80% of patients and of the platelet count ($< 400 \times 10^9/L$) in 55% of patients [14]. Nonetheless, thrombosis and the potential oncogenicity of prolonged exposure to HU in young patients are of concern. It is now well established that IFN- α 2b can control erythrocytosis or thrombocytosis in the majority of patients with PV or ET (the usual dose is three million units SC three times-a-week) [15].

The discovery of the JAK2V617F mutation introduced a new marker for evaluating treatment efficacy in MPDs. A decrease in JAK2V617F allele burden after IFN- α 2b treatment [8,16] has also been reported. Two recent studies of IFN α in PV and ET reported hematologic remissions of 80% accompanied by decreases in JAK2V617F allele burden (complete molecular remission rate of 5–10%) [17,18]. In the study, we found that for ET patients with positive JAK2V617F mutation, IFN α -2b has greater advantage

in OHR than negative JAK2V617F mutation. For PV patients with JAK2V617F⁺ mutation, IFN α -2b has no obvious therapy superiority to HU in OHR, but IFN α -2b regimen gains are more satisfactory OMR than HU. As we know, phlebotomy correlates with activity of PV. We indicated that IFN significantly decreased the phlebotomy rate than HU for MPDs patients with OHR than HU.

Because the survival rate in strictly WHO-defined PV and ET is near-normal and the 10-year risk of AML or myelofibrosis is less than 5% and 1% [4,19], it would be inappropriate to suggest that any current treatment could modify the natural history of the disease. We show that IFN α -2b has no superiority in five-year OS for MPDs with JAK2V617F⁺ because of the high survival rate for MPDs. Despite the neoplastic nature of MPDs, it is advisable to distinguish these disorders from leukemia, with this especially applying to PV and ET, associated with a life expectancy not substantially different from that of the general population [20]. However, in the study, ET patients with JAK2V617F⁺ demonstrate a definite advantage over IFN α -2b than JAK2V617F⁻ in five-year PFS. IFN α -2b is also superior to HU in five-year PFS for PV patients with JAK2V617F⁺. In PV group, the hematologic response had no more PFS advantages than patients who gained OHR, but molecular response is more important to progression of disease. Meanwhile, IFN α -2b also contributed to improving vasomotor symptom improvements to MPDs, especially significantly decreasing the incidence of distal paresthesias and erythromelalgia than HU. Moreover, IFN was a greater help in relieving vasomotor-related headaches in ET than in PV.

In the study, HU is not a drug without side effects; severe toxicity (Grade three or four) characterized by leucopenia and thrombocytopenia was observed in 1–5% of patients with PV. IFN α -2b is a nonleukemogenic agent active in MPDs [21]. Some patients initially suffered influenza-like symptoms to a greater or lesser degree, but no patients have side effects significant enough to give up the IFN α therapy. Silver et al. reported that 100% of patients treated with rIFN α were thrombosis-free survival. In the study, a total of 7/187 (3.7%) MPDs patients with IFN treatment died; six died of thrombosis-related disease, and one died in an unexpected accident. Differing

from Silver et al.'s findings, our subjects were advanced MPDs with extreme thrombocytosis risk or thrombosis history. Compared with the myelosuppressive drugs that have been used in treating ET or PV, IFN α -2b therapy had no severe hematological AEs observed in patients with PV or ET.

In conclusion, IFN α -2b has positive activities against phlebotomy and vasomotor symptoms, and it can improve JAK2V617F⁺ mutation-related molecular response and progression-free survival. We suggest that IFN has a unique role in the treatment of the myeloproliferative disorders, particularly PV or ET with JAK2V617F⁺ mutation.

Conflict of interest statement

All authors discussed the results and commented on the manuscript. I would like to declare on behalf of my co-authors that this work is original research that has not been published previously and is not under consideration for publication elsewhere, either in part or in whole. All the authors listed have approved the manuscript that is enclosed.

Acknowledgments

We would like to thank all the patients and the clinicians from the participating sites. This work was supported by grants from the Natural Science Foundation of Inner Mongolia (2013MS1157) and Inner Mongolia Autonomous Region Health and Family Planning health research projects (201302059).

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